

## AMENDMENT NO. 3 JUNE 2021

### TO

## IS 9271 : 2004 UNPLASTICIZED POLYVINYL CHLORIDE (UPVC) SINGLE WALL CORRUGATED PIPES FOR DRAINAGE — SPECIFICATION

( *First Revision* )

( *Page 1, clause 2* ) — Substitute the following for the existing clause:

‘The following standards contain provisions which, through reference in this text, constitute provision of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate, the possibility of applying the most recent editions of the standards.

| <i>IS No.</i>                    | <i>Title</i>                                                                                                                                                          |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4558 : 1995                      | Under-drainage of lined canals — Code of practice ( <i>second revision</i> )                                                                                          |
| 4669 : 1968                      | Methods of test for polyvinyl chloride resins                                                                                                                         |
| 4905 : 2015/<br>ISO 24153 : 2009 | Random sampling and randomization procedures ( <i>first revision</i> )                                                                                                |
| 4985 : 2021                      | Unplasticized PVC pipes for water supplies — Specification ( <i>fourth revision</i> )                                                                                 |
| 10151 : 2019                     | Polyvinyl chloride (PVC) and its copolymers for its safe use in contact with foodstuffs, pharmaceuticals and drinking water — Specification ( <i>first revision</i> ) |
| 12235                            | Thermoplastics pipes and fittings — Methods of test ( <i>first revision</i> )                                                                                         |

**Price Group 1**

**Amendment No. 3 to IS 9271 : 2004**

| <i>IS No.</i>                                | <i>Title</i>                                                                                                                                                           |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Part 4) : 2004                              | Determining the detrimental effect on the composition of water                                                                                                         |
| (Part 10) : 2004                             | Determination of organotin as tin aqueous solution                                                                                                                     |
| IS 13360                                     | Plastics — Methods of Testing                                                                                                                                          |
| (Part 3/Sec 10) : 2016/<br>ISO 1183-1 : 2012 | Physical and dimensional properties, Section 10<br>Determination of density of non-cellular plastics — Immersion method, liquid pycnometer method and titration method |
| (Part 3/Sec 11) : 2016/<br>ISO 1183-2 : 2004 | Physical and dimensional properties, Section 11<br>Determination of density of non-cellular plastics — Density gradient column method                                  |
| (Part 3/Sec 12) : 2016/<br>ISO 1183-3 : 1999 | Physical and dimensional properties, Section 12<br>Determination of density of non-cellular plastics — Gas pycnometer method'                                          |

(Page 3, clause 8, Title) — Substitute '**PHYSICAL AND CHEMICAL CHARACTERISTICS**' for '**PHYSICAL REQUIREMENTS**'.

(Page 3, clause 8.2) — Substitute 'IS 13360 (Part 3/Sec 10) or IS 13360 (Part 3/Sec 11) or IS 13360 (Part 3/Sec 12)' for 'IS 13360 (Part 3/Sec 1)'.

(Page 3, clause 8.7) — Insert the following clause at the end:

**'8.8 Effect on Water**

The pipes shall not have any detrimental effect on the composition of water flowing through them. When tested by the method described in IS 12235 (Part 4) and IS 12235 (Part 10), the quantities of lead, dialkyl tin C4 and higher homologues (measured as tin), and any other toxic substances extracted from the internal walls of the pipes shall not exceed the concentrations as specified in **10.3** of IS 4985 and meet the other requirements given in **10.3.1** of IS 4985.

NOTE — Implementation of the phase out programme for use of lead stabilizers in PVC pipe and fitting manufacturing of the Government of India shall be borne in mind.'

(Page 3, clause 9.1, Title) — Substitute ‘**Acceptance Test**’ for ‘**Lot**’.

(Page 3, clause 9.1.1) — Substitute the following for the existing:

**‘9.1.1 Lot**

In a single consignment, all the drainage pipes of the same type, same size, same length and manufactured under similar conditions of manufacture shall constitute of a lot.’

(Pages 3 and 4, clauses 9.1.2, 9.1.3 and 9.1.3.1 and Table 4) — Substitute ‘9.1.1.2’ for ‘9.1.2’, ‘9.1.1.3’ for ‘9.1.3’ and ‘9.1.1.3.1’ for ‘9.1.3.1’.

(Page 4, clause 9.2) — Substitute ‘9.1.2 Number of Test and Criteria for Conformity’ for ‘9.2 Number of Test and Criteria for Conformity’.

(Page 4, clauses 9.2, 9.2.1, 9.2.2, 9.2.2.1 and 9.2.3 and Tables 4 and 5) — Substitute ‘9.1.2.1’ for ‘9.2.1’, ‘9.1.2.2’ for ‘9.2.2’, ‘9.1.2.2.1’ for ‘9.2.2.1’, and ‘9.1.2.3’ for ‘9.2.3’.

[Page 4, clause 9.2.1 (now renumbered as 9.1.2.1)] — Substitute ‘col 4’ for ‘col 3’, ‘col 6’ for ‘col 5’, ‘col 7’ for ‘col 6’ and ‘col 6 and 7’ for ‘col 5 and 6’.

[Page 4, clause 9.2.3 (now renumbered as 9.1.2.3)] — Insert the following new clause:

**‘9.2 Type Tests**

**9.2.1** Type tests are intended to prove the suitability and performance of a new composition or a new size of pipe. Such tests, therefore, need to be applied only when a change is made in polymer composition or when a new size of pipe is to be introduced. Type tests for compliance with effect on water test as specified in 8.8 shall be carried out.

**9.2.2** For effect on water test, the manufacturer or the supplier shall furnish to the testing authority three samples of the smallest size of pipe taken from each machine (selected preferably from a regular production lot).

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**9.2.2.1** Three samples so selected shall be tested for compliance with the requirements for effect on water as given in **8.8**.

**9.2.2.2** If all three samples pass the requirements for effect on water, the type test of the pipe under consideration shall be considered to be eligible for approval, which shall be normally valid for a period of one year.

**9.2.2.3** In case any of the samples fails in this test, the testing authority, at its discretion, may call for fresh samples not exceeding the original number, and subject them to the test for effect on water. If, in the repeat test, no single failure occurs, the type of pipe under consideration shall be considered eligible for type approval. If any of the samples fails in the repeat test, the type of pipe shall not be approved. The manufacturer or the supplier may be asked to improve the design and resubmit the product for type approval.

**9.2.2.4** At the end of the validity period (normally one year) or earlier, if necessary, the testing authority may call for fresh samples for effect on water test for the purpose of type approval.’

(Page 4, *Clause 10.1*) — Insert the following clause at the end:

**‘10.1.1** The lot number/batch number shall include the details of production in the following manner:

| Year | Month | Day | Machine No. | Shift |
|------|-------|-----|-------------|-------|
| xxxx | xx    | xx  | xxx         | x     |

(Page 4, *clause 10.2*) — Substitute the following for the existing:

**‘10.2 BIS Certification Marking**

The drainage pipes conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the drainage pipes may be marked with the Standard Mark.’